Complex Bash

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This handout is intended as a resource for practicing unconventional and advanced problems in complex bashing. I have also noticed that several handouts have problems that are bash*able*, but no one in their right mind would ever bash them, so I have tried to include problems where I believe a bash is the most efficient approach.

1 Techniques

1.1 Warm up

Problem 1

Let ABC be a triangle with incenter I and circumcircle Ω. Let D, E, F be the midpoints of the arcs $\overset{\frown}{BC}, \overset{\frown}{CA}, \overset{\frown}{AB}$ of Ω not containing A, B, C respectively. Let D' be the point of Ω diametrically opposite to D. Show that I, D' and the midpoint M of EF lie on a line.

Proof. Notice $a = x^2$, $b = y^2$, $c = z^2$, $d = -yz$, $e = -zx$, $f = -xy$, $j = -(xy + yz + zx)$, $d' = yz$, and $m = \frac{-xy-yz}{2}$. Then,

$$\frac{2yz + xy + xz}{\frac{2yz+xy+xz}{2}} = 2 \in \mathbb{R}$$

□

Problem 2

The foot of the altitude from vertex A of acute triangle ABC is T_A. The ray drawn from A through the circumcentre O intersects BC at R_A. Let the midpoint of AR_A be F_A. Define T_B, R_B, F_B, T_C,

R_C, F_C similarly. Prove that T_AF_A, T_BF_B, T_CF_C are concurrent.

Proof. Let (ABC) be the unit circle. $T_A = \frac{1}{2}\left(a+b+c-\frac{bc}{a}\right)$.

$R_A = BC \cap AA' = \frac{a^2(b+c)}{a^2+bc}$.

$F_A = \frac{a+R_A}{2} = \frac{1}{2}\left(\frac{a(a^2+bc)+a^2(b+c)}{a^2+bc}\right)$.

Since

$$\frac{n_9 - F_A}{n_9 - T_a} = \frac{a(b+c)}{a^2+bc} = \overline{\frac{a(b+c)}{a^2+bc}}$$

T_A, F_A, N_9 are collinear. By symmetry, T_AF_A, T_BF_B, T_CF_C concur at N_9. □

1.2 Homothety

Problem 3

In triangle ABC with $CA = CB$, point E lies on the circumcircle of ABC such that $\angle ECB = 90°$. The line through E parallel to CB intersects CA in F and AB in G. Prove that the centre of the circumcircle of triangle EGB lies on the circumcircle of triangle ECF.

Proof. Let O and P be the circumcentres of $\triangle ABC$ and $\triangle EGB$ respectively, and let EF intersect (ABC) again at T.

Clearly, E and T are the antipodes of B and C respectively.

Notice that $\measuredangle FEO = \measuredangle TEB = \measuredangle TCB = \measuredangle ACT = \measuredangle FCO$, so C, E, F, O are concyclic.

Let's employ complex numbers with respect to (ABC), with $a = \frac{c^2}{b}$, $e = -b$, $t = -c$.

The intersection formula gives $g = \frac{b^2+bc+2c^2}{b-c}$. Shifting by b and taking a homothety with scale factor $\frac{1}{2}$ gives $b' = b$, $g' = \frac{b^2+c^2}{b-c}$, $e' = 0$.

Now, using the circumcircle formula and shifting back gives $p = \frac{b(b+c)}{b-c}$. It suffices to prove C, E, O, P are concyclic $\iff \left(\frac{c-p}{e-p}\right) \times \frac{e}{c} = \frac{b}{c} + \frac{c}{b} \in \mathbb{R}$ Which is evidently true. □

Problem 4

Let ABC be an acute triangle with incentre I and $AB \neq AC$. Let lines BI and CI intersect the circumcircle of ABC at $P \neq B$ and $Q \neq C$, respectively. Consider points R and S such that $AQRB$ and $ACSP$ are parallelograms (with $AQ \parallel RB, AB \parallel QR, AC \parallel SP$, and $AP \parallel CS$). Let T be the point of intersection of lines RB and SC. Prove that points R, S, T, and I are

concyclic.

Proof. Notice that $\measuredangle RTS = \measuredangle QAP$, so it suffices to prove $\measuredangle RIS = \measuredangle QAP$.

Now, work on the complex plane with $a = x^2, b = y^2, c = z^2$. So, $p = -xz, q = -xy, j = -xy - yz - zx, r = y^2 - xy - x^2, s = z^2 - xz - x^2$.

$$\left(\frac{r-j}{s-j}\right)\left(\frac{a-p}{a-q}\right) \in \mathbb{R}$$

$$\iff \frac{(x+y)(y-x+z)}{(x+z)(y-x+z)} \times \frac{x+z}{x+y} = 1 \in \mathbb{R}$$

which is trivially true. □

Problem 5

In triangle ABC, consider points A_1, A_2 on line BC such that A_1, B, C, A_2 are in that order and $A_1B = AC$ and $CA_2 = AB$. Similarly consider points B_1, B_2 on line AC, and C_1, C_2 on line AB. Prove that the orthocentres of triangles $A_1B_1C_1$ and $A_2B_2C_2$ are equidistant from the circumcentre of ABC.

Proof. Let DEF be the intouch triangle of $\triangle ABC$ with centre I, and let H_1, H_2, H be the orthocentres of $\triangle A_1B_1C_1, \triangle A_2B_2C_2, \triangle DEF$ respectively. Let O be the circumcentre of ABC.

Notice that $A_1, A_2, B_1, B_2, C_1, C_2$ are concyclic with centre I, since $DI = EI = FI$ and $DA_1 = DA_2 = EB_1 = EB_2 = FC_1 = FC_2 = s$, where s is the semiperimeter of $\triangle ABC$

H is the midpoint of $\overline{H_1H_2}$

Proof. Let $A_1A_2B_1B_2C_1C_2$ be the unit circle with $A_1 = a$, $B_1 = b$, $C_1 = c$, $A_2 = \omega a$, $B_2 = \omega b$, $C_2 = \omega c$ since $A_1A_2 = B_1B_2 = C_1C_2$.

D, E, F are the midpoints of A_1A_2, B_1B_2, C_1C_2 respectively, so $d = \frac{a(1+\omega)}{2}$ and so on.

Taking a homothety with scale factor $\frac{2}{1+\omega}$ and scaling back gives $h = \frac{(1+\omega)(a+b+c)}{2}$, which is precisely the midpoint of $\overline{H_1H_2}$. □

Additionally, $HI \perp H_1H_2$ since $\frac{\frac{(1+\omega)(a+b+c)}{2}}{(1-\omega)(a+b+c)} = \frac{1+\omega}{2(1-\omega)}$ is purely imaginary.

To finish, note that the inversion with respect to the incircle takes O to the nine-point center of $\triangle DEF$, which lies on IH. □

Problem 6

Let ΔABC be an acute angled scalene triangle with circumcircle ω. Let O and H be the circumcenter and orthocenter of ΔABC, respectively. Let E, F and Q be points on segments AB, AC and ω, respectively, such that

$$\angle BHE = \angle CHF = \angle AQH = 90^\circ.$$

Prove that OQ and AH intersect on the circumcircle of ΔAEF.

Proof. Let the antipode of Q be Q', and let AH meet ω again at A'. Let T be $QQ' \cap AA'$. We need T to lie on (AEF).

Let (ABC) be the unit circle, with $O = 0$ and $h = a + b + c$.

$$\frac{e-a}{e-b} = \frac{\overline{e}a - 1}{\overline{e}b - 1} \times \frac{b}{a} \implies \overline{e} = \frac{a+b-e}{ab}$$

$$\frac{h-b}{h-e} = \frac{a+c}{a+b+c-e} = -\frac{b(a+c)}{ab+ec} \implies e = \frac{b(2a+b+c)}{b-c}$$

Similarly $f = \frac{c(2a+b+c)}{c-b}$.

Now,

$$\frac{q-a}{q-a-b-c} = \frac{bc(q-a)}{abc - q(ab+bc+ca)} \implies q = \frac{2abc + b^2c + bc^2}{ab + 2bc + ac}$$

$$a' = \frac{a-a-b-c}{\frac{ab+bc+ca}{bc} - 1} = -\frac{bc}{a}$$

To compute T, notice

$$t = \frac{qq'(a+a') - aa'(q+q')}{qq' - aa'} = \frac{q^2(a^2-bc)}{a(q^2-bc)} = -\frac{bc(2a+b+c)^2}{a(b-c)^2}$$

Take a homothey with scale factor $\frac{b-c}{2a+b+c}$ to get $e = b$, $f = -c$, $a = \frac{a(b-c)}{2a+b+c}$, $t = -\frac{bc(2a+b+c)}{a(b-c)}$.

$$\frac{a-f}{e-f} = \frac{a+c}{2a+b+c}$$

and

$$\frac{a-t}{e-t} = \frac{a^2(b+c)+4abc+bc(b+c)}{b(2a+b+c)(a+c)}$$

So,

$$\frac{a-f}{e-f} \times \frac{e-t}{a-t} = \frac{b(a+c)^2}{a^2(b+c)+4abc+bc(b+c)}$$

which is equal to its conjugate, so we are done. □ □

1.3 Inversion

Problem 7

Let I be the incentre of acute triangle ABC with $AB \neq AC$. The incircle ω of ABC is tangent to sides BC, CA, and AB at D, E, and F, respectively. The line through D perpendicular to EF meets ω at R. Line AR meets ω again at P. The circumcircles of triangle PCE and PBF meet again at Q.

Prove that lines DI and PQ meet on the line through A perpendicular to AI.

Proof. Let D' be the antipode of D on ω, and let T be a point on DI such that $AT \parallel EF$. It suffices to prove P, Q, T are collinear.

Invert with respect to the incircle. We now have the following problem, removing *s:

Let triangle D, E, F have midpoints A, B, C. Define P, Q, R, D' similar to the original problem, and let $Q = (BFP) \cap (CEP)$. Prove that Q lies on the nine-point circle Ω of ADD'.

P, A, D' collinear

Proof. Work on the complex plane. Let $D = d$, $E = e$ and $F = f$ so $r = \frac{-ef}{d}$.

Now, we compute

$$p = \frac{r-a}{r\overline{a}-1} = \frac{2def+ef(e+f)}{d(e+f)+2ef}$$

So, we obtain

$$\frac{1}{2} \times \frac{d'-p}{d'-a} = \frac{(d^2+ef)(e+f)+4def}{(2d+e+f)(2ef+de+df)}$$

which is equal to its conjugate. □

Additionally, $Q \in BC$ since

$$\measuredangle PQB = \measuredangle PFD = \measuredangle PEC = \measuredangle PQC$$

To finish, notice that the midpoint M of AD and BC lies on Ω, and

$$\measuredangle MIP = \measuredangle D'PI = \measuredangle ID'P = \measuredangle DEP = \measuredangle MQP$$

□

Problem 8

Let ABC be an acute-angled triangle with circumcircle ω and circumcentre O. Points $D \neq B$ and $E \neq C$ lie on ω such that $BD \perp AC$ and $CE \perp AB$. Let CO meet AB at X, and BO meet AC at Y.

Prove that the circumcircles of triangles BXD and CYE have an intersection lie on line AO.

Proof. We claim the intersection is $P = AO \cap (BOC)$. For this, we use complex numbers with unit circle (ABC).

Notice that inversion at O sends P to $AO \cap BC$, which is just

$$\frac{bc(a-a) + a^2(b+c)}{bc + a^2} = \frac{a^2(b+c)}{a^2 + bc}$$

by the intersection formula.

Inverting back gives us

$$p = \left(\frac{\overline{a^2(b+c)}}{a^2 + bc}\right)^{-1} = \frac{a^2 + bc}{b+c}$$

Additionally,

$$\frac{b-d}{a-c} = \frac{\frac{1}{b} - \frac{1}{d}}{\frac{1}{a} - \frac{1}{c}} \implies d = -\frac{ac}{b}$$

and

$$x = \frac{ab(c-c) - (-c^2)(a+b)}{ab - (-c^2)} = \frac{c^2(a+b)}{ab + c^2}$$

Finally,

$$\frac{(b-x)(d-t)}{(d-x)(b-t)} = \frac{a(b^2-c^2)}{c(b^2-a^2)}$$

Which is equal to its conjugate. So, we are done by symmetry. □

2 Practise Problems

I am going to be omitting the easy section, since this is *not* meant to be a first course in complex bashing.

2.1 Medium

1 Let ABC be a triangle with $AB < AC < BC$. Let the incenter and incircle of triangle ABC be I and ω, respectively. Let X be the point on line BC different from C such that the line through X parallel to AC is tangent to ω. Similarly, let Y be the point on line BC different from B such that the line through Y parallel to AB is tangent to ω. Let AI intersect the circumcircle of triangle ABC at $P \neq A$. Let K and L be the midpoints of AC and AB, respectively.

Prove that $\angle KIL + \angle YPX = 180^\circ$.

2 Let ABC be a triangle and D, E, F be the midpoints of arcs BC, CA, AB on the circumcircle. Line ℓ_a passes through the feet of the perpendiculars from A to DB and DC. Line m_a passes through the feet of the perpendiculars from D to AB and AC. Let A_1 denote the intersection of lines ℓ_a and m_a. Define points B_1 and C_1 similarly. Prove that triangle DEF and $A_1B_1C_1$ are similar to each other.

3 Quadrilateral $APBQ$ is inscribed in circle ω with $\angle P = \angle Q = 90^\circ$ and $AP = AQ < BP$. Let X be a variable point on segment $\overline{PQ}$. Line AX meets ω again at S (other than A). Point T lies on arc AQB of ω such that $\overline{XT}$ is perpendicular to $\overline{AX}$. Let M denote the midpoint of chord $\overline{ST}$. As X varies on segment $\overline{PQ}$, show that M moves along a circle.

4 Let ABC be a non-isosceles triangle with incenter I whose incircle is tangent to $\overline{BC}$, $\overline{CA}$, $\overline{AB}$ at D, E, F, respectively. Denote by M the midpoint of $\overline{BC}$. Let Q be a point on the incircle such that $\angle AQD = 90^\circ$. Let P be the point inside the triangle on line AI for which $MD = MP$. Prove that either $\angle PQE = 90^\circ$ or $\angle PQF = 90^\circ$.

5 Let O be the circumcenter of an acute triangle ABC. Line OA intersects the altitudes of ABC through B and C at P and Q, respectively. The altitudes meet at H. Prove that the circumcenter of triangle PQH lies on a median of triangle ABC.

2.2 Hard

1 Let ABC be an acute, scalene triangle with orthocentre H. Let ℓ_a be the line through the reflection of B with respect to CH and the reflection of C with respect to BH. Lines ℓ_b and ℓ_c are defined similarly. Suppose lines ℓ_a, ℓ_b, and ℓ_c determine a triangle $\mathcal{T}$.

Prove that the orthocentre of $\mathcal{T}$, the circumcentre of $\mathcal{T}$, and H are collinear.

2 Let D be an interior point of the acute triangle ABC with $AB > AC$ so that $\angle DAB = \angle CAD$. The point E on the segment AC satisfies $\angle ADE = \angle BCD$, the point F on the segment AB satisfies $\angle FDA = \angle DBC$, and the point X on the line AC satisfies $CX = BX$. Let O_1 and O_2 be the circumcenters of the triangles ADC and EXD, respectively. Prove that the lines BC, EF, and O_1O_2 are concurrent.

3 Let ABC be an acute-angled triangle with $AB < AC$. Let Ω be the circumcircle of ABC. Let S be the midpoint of the arc CB of Ω containing A. The perpendicular from A to BC meets BS at D and meets Ω again at $E \neq A$. The line through D parallel to BC meets line BE at L. Denote the circumcircle of triangle BDL by ω. Let ω meet Ω again at $P \neq B$. Prove that the line tangent to ω at P meets line BS on the internal angle bisector of $\angle BAC$.

4 Let I be the incenter of acute triangle ABC with $AB \neq AC$. The incircle ω of ABC is tangent to sides BC, CA, and AB at D, E, and F, respectively. The line through D perpendicular to EF meets ω at R. Line AR meets ω again at P. The circumcircles of triangle PCE and PBF meet again at Q.

Prove that lines DI and PQ meet on the line through A perpendicular to AI.

5 Let ABC be an acute triangle with circumcircle Γ. Let ℓ be a tangent line to Γ, and let ℓ_a, ℓ_b and ℓ_c be the lines obtained by reflecting ℓ in the lines BC, CA and AB, respectively. Show that the circumcircle of the triangle determined by the lines ℓ_a, ℓ_b and ℓ_c is tangent to the circle Γ.

6 Let ABC be an acute triangle. Let M be the midpoint of side BC, and let E and F be the feet of the altitudes from B and C, respectively. Suppose that the common external tangents

to the circumcircles of triangles BME and CMF intersect at a point K, and that K lies on the circumcircle of ABC. Prove that line AK is perpendicular to line BC.